



A1 Water

A · Unity and diversity · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Water is a polar molecule: the oxygen atom carries a slight negative charge and each hydrogen a slight positive charge, allowing hydrogen bonds to form between adjacent water molecules. These hydrogen bonds give water cohesion, high specific heat capacity, high latent heat of vaporization, and make it an excellent solvent for polar and charged solutes. These properties underlie thermoregulation, long-distance water transport in plants, and the aqueous environment needed for metabolic reactions.

Worked example: follow the reasoning

Explaining cohesion-tension in a tall tree: (1) Water evaporates from mesophyll cell walls in the leaf (transpiration), lowering water potential in the leaf xylem. (2) Hydrogen bonds link water molecules into a continuous, unbroken column inside the narrow xylem vessels. (3) This cohesion allows a pulling force (tension) generated at the leaf to be transmitted all the way down the column to the roots. (4) Adhesion of water to the lignified xylem walls resists the column collapsing or being pulled sideways. (5) As water is pulled up, more is drawn into the root by osmosis. Conclusion: cohesion between water molecules, powered by evaporation, moves water up to the top of a tree without any direct expenditure of metabolic energy by the plant.

Quick check · multiple choice

Which property of water is a direct consequence of hydrogen bonding between water molecules?

- A Its ability to dissolve non-polar gases readily
- B Its unusually high specific heat capacity
- C Its low boiling point compared with similarly sized molecules
- D Its low surface tension

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student sweats 45 g of water during a workout, all of which evaporates from the skin. The latent heat of vaporization of water is approximately 2260 J g^{-1} . [6]
- (a) Calculate the heat energy removed from the body by this evaporation. [2]
 - (b) Explain, in terms of hydrogen bonding, why evaporation of sweat is an effective cooling mechanism. [2]
 - (c) Water also has a high specific heat capacity. Explain why this property helps organisms living in aquatic habitats resist rapid temperature change. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Heat = mass $\times$ latent heat = $45 \text{ g} \times 2260 \text{ J g}^{-1} = 101\,700 \text{ J} \approx 1.02 \times 10^5 \text{ J}$ (101.7 kJ) [1 mark for correct setup, 1 for correct value with units].

(b) Evaporation requires breaking many hydrogen bonds between liquid water molecules; this energy is drawn from the body/skin surface as heat, so the skin loses a large amount of heat energy for a relatively small mass of water evaporated, cooling the body.

(c) High specific heat capacity means a large amount of energy must be absorbed or released to change the water's temperature (because many hydrogen bonds must be broken/reformed); this buffers aquatic organisms against rapid, large temperature fluctuations in their environment, providing a thermally stable habitat.

Quick check answer

B. Breaking many hydrogen bonds to raise the temperature of water requires a large energy input, giving water a high specific heat capacity.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.